

JAYANTH DASAMANTHARAO

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway, EliteUS Software Solutions

Aug 2024 - Present | Piscataway, NJ

- Engineered an AI chatbot using **NLP, AWS Bedrock** and Streamlit for system instructions and real-time data-driven decision-making. Integrated **Language Models** for context-aware responses, scalability.
- Integrated vector databases (**Pinecone, Chroma**) and **LangChain** for **retrieval-augmented generation (RAG)**, driving insight-based responses and improved campaign optimization.
- Tech Stack: Python, AWS Bedrock, Azure, OpenAI, LangChain, Chroma, VectorDB, PyMuPDF, Prompting.

Data Science Intern, Harvest Software Solutions, LLC

June 2023 - Sept 2023 | Jacksonville, Florida

- Optimized real-time ad campaigns by deploying **H2O AutoML and XGBoost models**, conducting **A/B tests** to measure and enhance marketing performance, achieving boost in predictive accuracy, ad spend efficiency.
- Leveraged **FastAPI** to deploy NLP models with Hugging Face Transformers for customer sentiment and topic analysis on large-scale data, providing actionable insights on engagement using Python.

Application Development Associate, Accenture Solutions Pvt. Ltd.

June 2021 - Aug 2022 | Hyderabad, India

Database Analyst, Regeneron Pharmaceuticals Inc. - Client Data Analysis & Warehousing

- Optimized **ETL** workflows and implemented **automated data pipelines in Python**, enhancing data processing efficiency.
- Processed large-scale datasets, leveraging **PySpark, SQL, AWS** like **Athena and Redshift** to process and analyze data efficiently. Created interactive dashboards, visualized using **Tableau, PowerBI** to support data-driven decision-making.
- Transformed data within a **relational database environment**, increasing system reliability through **DDL optimization**, data integration, performance tuning, ensuring stable data warehousing and reporting operations using **SQL Workbench**.

Research Lab, Andhra University

January 2021 - June 2021 | Visakhapatnam, India

Self Driving Cars Using Image Processing and Deep Learning

- Engineered **Convolutional Neural Networks (CNNs)** for autonomous lane detection and object recognition using python.
- Leveraged **YOLO and Faster R-CNN** architectures to optimize real-time detection and classification accuracy.
- Enhanced model generalization through **image pre-processing** techniques like **Gaussian blurring and color space conversion**, and integrated **multi-sensor data** from **radar, lidar, and cameras** to improve navigational precision.
- Developed a real-time simulation environment using **Unity3D and a Flask backend** for dynamic parameter monitoring and system validation, enabling high-fidelity testing of autonomous driving functionalities.

TECHNICAL SKILLS

Programming: SQL, Python, R, Scala, Stata. **Libraries:** Pytorch, Tensorflow, Seaborn, Plotly, Matplotlib, Shiny, D3.js.
Data Analytics: Tableau, Power BI, Data Wrangling, Data Pipelines, EDA, ETL, A/B Testing, Google Analytics.
Machine Learning: Regression, Classification, Clustering, Feature Engineering, Predictive Analytics, Anomaly Detection.
Statistical Analysis: Hypothesis testing, A/B testing, Bayesian Inference, Multivariate testing, T-test, Stochastic process.
Deep Learning: Neural Networks, Computer Vision, AI, NLP, BERT, GANs, LLMs, LLaMA, LSTM, Data Mining.
Database & Cloud: AWS, Azure, GCP, Data Warehousing, RDBMS, MongoDB, BigQuery, Langchain, Vectordb, Pinecone.
Data Engineering: PostgreSQL, Tableau prep, Kafka, Apache Spark, Git, Data Warehousing, Databricks, Hadoop, Docker.

EDUCATION

Rutgers University, Masters in Data Science

Sept 2022 - May 2024 | New Brunswick, NJ

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** [Portfolio](#).

Andhra University, B.Tech in Electrical and Electronics Engineering

2017 - 2021 | Visakhapatnam, India

RESEARCH PROJECTS

Bayesian Classification - [Bayesian A/B testing, Bayesian Networks, PyMC3, Informative Prior Distributions]
Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.

Thank you for considering my application. I'm available to start working immediately.